Tech Dessert #10: Digital and Internet-Linked Operating

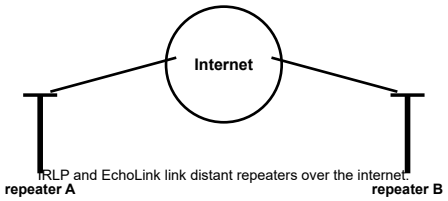
Companion reference for meeting 10 -- operating activities, internet-linked radio, and the digital modes a Technician will meet.

Groups T8C + T8D · 23 pool questions · Self-study handout · Practice exams at eham.net/exams

This Dessert covers the activities and digital modes that make modern ham radio fun – contests, fox hunts, satellites of the internet kind, and the data modes. It is a reference sheet: read the tables, study the diagram, then drill the question bank.

Operating Activities

Topic	What to know	Q
Locating a noise or jamming source	**Radio direction finding.**	T8C01
Useful gear for a hidden transmitter hunt	A **directional antenna**.	T8C02
Contesting	Contacting **as many stations as possible** in a set period.	T8C03
Good contest practice	Send only the **minimum information** needed to identify and exchange.	T8C04
A grid locator	A **letter-number designator** for a geographic location.	T8C05
IRLP (Internet Radio Linking Project)	Connects amateur systems, such as repeaters, **over the internet**.	T8C08
Accessing IRLP over the air	Using **DTMF** tones.	T8C06
VoIP (Voice Over Internet Protocol)	Delivering voice over the internet using **digital techniques**.	T8C07
EchoLink	Lets you transmit through a repeater **without using a radio** to start it.	T8C09
Before using EchoLink	**Register your call sign** and provide proof of license.	T8C10
A gateway	An amateur station that **connects other stations to the internet**.	T8C11



Internet linking lets two repeaters far apart act as one – you talk through your local repeater and reach the other.

Digital Modes

Topic	What to know	Q
What counts as a digital mode	Packet radio, IEEE 802.11 (Wi-Fi), and FT8 – **all of these**.	T8D01
FT8	A digital mode that works at very **low signal-to-noise** – weak signals.	T8D02
CW	Another name for a **Morse code** transmission.	T8D09
PSK	**Phase Shift Keying.**	T8D06
DMR	**Time-multiplexes two digital voice signals** on one 12.5 kHz channel.	T8D07
What data APRS can carry	GPS position, text messages, and weather data – **all of these**.	T8D03
An APRS application	Real-time tactical digital communications with a **map of station locations**.	T8D05
NTSC	An **analog fast-scan color TV** signal.	T8D04
What packet radio includes	A checksum, a header with the destination call sign, and ARQ – **all of these**.	T8D08
ARQ	Error correction: the receiver detects errors and **requests a retransmission**.	T8D11
What WSJT-X software supports	Earth-Moon-Earth, weak-signal beacons, and meteor scatter – **all of these**.	T8D10
An amateur mesh network	A data network built on **commercial Wi-Fi equipment with modified firmware**.	T8D12

Remember: radio links up with the internet in several ways – **IRLP** and **EchoLink** connect distant repeaters, and a **gateway** connects stations to the internet. The digital modes each have a job: **FT8** for weak signals, **APRS** for position and tactical data, **DMR** and **PSK** for digital voice and text. And good fun lives here too – **contesting**, **fox hunts** (radio direction finding), and **grid locators**.

Dessert 10 Question Bank (T8C + T8D)

T8C01. Which of the following methods is used to locate sources of noise interference or jamming?
A. Echolocation
B. Doppler radar
C. Radio direction finding
D. Phase locking

T8C02. Which of these items would be useful for a hidden transmitter hunt?
A. Calibrated SWR meter
B. A directional antenna
C. A directional wattmeter
D. All these choices are correct

T8C03. What operating activity involves contacting as many stations as possible during a specified period?
A. Simulated emergency exercises
B. Net operations
C. Hidden transmitter hunts
D. Contesting

T8C04. Which of the following is good practice when contacting another station in a contest?
A. Signing only the last two letters of your call if there are many other stations calling
B. Contacting the station twice to be sure that you are in his log
C. Sending only the minimum information needed for proper identification and the contest exchange
D. Adding “Please copy” before your exchange

T8C05. What is a grid locator?
A. A letter-number designator assigned to a geographic location
B. A letter-number designator assigned to an azimuth and elevation
C. An instrument for locating faults in power amplifiers
D. An instrument for radio direction finding

T8C06. How is over the air access to Internet Radio Linking Project (IRLP) nodes accomplished?
A. By obtaining a password that is sent via voice to the node
B. By using Dual-Tone Multi-Frequency (DTMF) signals
C. By entering the proper internet password
D. By using Continuous Tone-Coded Squelch System (CTCSS) tone codes

T8C07. What is Voice Over Internet Protocol (VoIP)?
A. A set of rules specifying how to identify your station when linked over the internet to another station
B. A technique employed to “spot” DX stations via the internet
C. A technique for measuring the modulation quality of a transmitter using remote sites monitored via the internet
D. A method of delivering voice communications over the internet using digital techniques

T8C08. What is the Internet Radio Linking Project (IRLP)?
A. A technique to connect amateur radio systems, such as repeaters, via the internet
B. A system for providing access to websites via amateur radio
C. A system for informing amateurs in real time of the frequency of active DX stations
D. A technique for measuring signal strength of an amateur transmitter via the internet

T8C09. Which of the following protocols enables an amateur station to transmit through a repeater without using a radio to initiate the transmission?
A. IRLP
B. D-STAR
C. DMR
D. EchoLink

T8C10. What is required before using the EchoLink system?
A. Complete the required EchoLink training
B. Purchase a license to use the EchoLink software
C. Register your call sign and provide proof of license
D. At least a General Class license

T8C11. What is an amateur radio station that connects other amateur stations to the internet?
A. A gateway
B. A repeater
C. A digipeater
D. A beacon

T8D01. Which of the following is a digital communications mode?
A. Packet radio
B. IEEE 802.11
C. FT8
D. All these choices are correct

T8D02. What is FT8?
A. A wideband FM voice mode
B. A digital mode capable of low signal-to-noise operation
C. An eight-channel multiplex mode for FM repeaters
D. A digital slow-scan TV mode with forward error correction and automatic color compensation

T8D03. What kind of data can be transmitted by APRS?
A. GPS position data
B. Text messages
C. Weather data
D. All these choices are correct

T8D04. What is meant by the term “NTSC?”
A. A digital transmission standard for encrypting data
B. A special mode for satellite uplink
C. An analog fast-scan color TV signal
D. A frame compression scheme for TV signals

T8D05. Which of the following is an application of APRS?
A. Providing real-time tactical digital communications in conjunction with a map showing the locations of stations
B. Automatically showing the number of packets transmitted via PACTOR during a specific time interval
C. Providing voice over internet connection between repeaters
D. Providing information on the number of stations signed into a repeater

T8D06. What does the abbreviation “PSK” mean?
A. Pulse Shift Keying
B. Phase Shift Keying
C. Packet Sampled Keying
D. Power Sampled Keying

T8D07. Which of the following describes DMR?
A. A technique for time-multiplexing two digital voice signals on a single 12.5 kHz repeater channel
B. An automatic position tracking mode for FM mobiles communicating through repeaters
C. An automatic computer logging technique for hands-off logging when communicating while operating a vehicle
D. A digital technique for transmitting on two repeater inputs simultaneously for automatic error correction

T8D08. Which of the following is included in packet radio transmissions?
A. A checksum that permits error detection
B. A header that contains the call sign of the station to which the information is being sent
C. Automatic repeat request in case of error
D. All these choices are correct

T8D09. What is CW?
A. A type of electromagnetic propagation
B. A digital mode used primarily on 2-meter FM
C. Error correction for digital transmission using code words
D. Another name for a Morse code transmission

T8D10. Which of the following operating activities is supported by digital mode software in the WSJT-X software suite?
A. Earth-Moon-Earth
B. Weak signal propagation beacons
C. Meteor scatter
D. All these choices are correct

T8D11. What is the role of ARQ in a transmission system?
A. A special transmission format limited to video signals
B. A system used to encrypt command signals to an amateur radio satellite
C. An error correction method in which the receiving station detects errors and sends a request for retransmission
D. A method of compressing data using autonomous reiterative Q codes prior to final encoding

T8D12. Which of the following best describes an amateur radio mesh network?
A. An amateur-radio data network using commercial Wi-Fi equipment with modified firmware
B. A wide-bandwidth digital voice mode employing DMR protocols
C. An amateur-radio satellite communications network using modified commercial satellite TV hardware
D. An internet linking protocol allowing communication through repeaters around the world

Answer Key

T8C01: C	T8C02: B	T8C03: D	T8C04: C	T8C05: A
T8C06: B	T8C07: D	T8C08: A	T8C09: D	T8C10: C
T8C11: A	T8D01: D	T8D02: B	T8D03: D	T8D04: C
T8D05: A	T8D06: B	T8D07: A	T8D08: D	T8D09: D
T8D10: D	T8D11: C	T8D12: A		